Data Management

IB9HP0: SmartCart Retail Analytics Database

Group Assignment Report

Master of Science in Business Analytics
2025-2026

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SmartCart Retail Analytics Database

1. Introduction

This report details the design, implementation, and analysis of a relational database for SmartCart, an omnichannel grocery and household retailer. The database is built using SQLite to store, manage, and analyse transactional and promotional data, ultimately driving data-informed marketing decisions.

2. Business Understanding

2.1 Company Overview

SmartCart is a medium-sized retail enterprise that operates various product categories such as food, household items, and electronic products, and conducts business through both online platforms and offline stores. The company's core operational activities include customer account management, promotion execution, product catalog maintenance, and sales transaction processing.

As a result, SmartCart generates substantial operational data. The company increasingly relies on this data to monitor sales performance and evaluate marketing effectiveness. However, it currently lacks a structured databasesystem to systematically manage and analyze operational data. The main business processes of SmartCart are presented in Appendix A.

2.2 Business Questions

The SmartCart database was designed to support business decision-making by addressing a specific managerial challenge. Although SmartCart regularly uses promotional campaigns to increase sales and encourage repeat purchases, the company currently lacks a structured way to assess whether these activities genuinely improve profitability.

Therefore, this analysis focuses on whether promotions create real profit rather than simply boosting sales volume. By examining transaction data across product categories, customer segments, and sales channels, the database provides a structured basis for evaluating whether promotions increase profit or erode margins.

3. Database Design

3.1 Mini-World Definition

This database models a retail scenario where customers purchase products through orders containing multiple items, each assigned to a category. Promotional campaigns may be applied to selected products or categories to influence purchasing behaviour.

The purpose of mini-world is to model the core data needed to answer the main business question: which promotional campaigns increase profit rather than only increasing sales volume. Focusing on the relationships between entities, the business can evaluate both sales performance and the financial impact of promotional strategies. The current conceptual diagram (As-is) of SmartCart is presented in Appendix B (Figure B1).

3.2 Logical Entity–Relationship Structure (Normalized Design)

To transition from the initial conceptual model to a functional relational database, the schema was normalised to Third Normal Form (3NF) to eliminate data redundancy. The logically structured Entity-Relationship (ER) design mirrors SmartCart's core retail operations across four distinct domains:

1. **Customer Management:** The Customer and Membership entities track user profiles and loyalty tier segmentation.
2. **Inventory:** The product catalogue is structured hierarchically via the Product and Category tables.
3. **Sales Transactions:** Purchasing events are captured through a parent-child structure, separating overarching invoice details (Orders) from granular line-item metrics (Order_Items).
4. **Marketing Campaigns:** The Promotion entity stores discount rules, supported by mapping tables to link these campaigns directly to target inventory.

The complete logical structure, including relational cardinalities, is visualised using Crow's Foot notation in Appendix B (Figure B2). Furthermore, a comprehensive breakdown of all entities, their primary functions, and their structural keys (Primary and Foreign Keys) is detailed in Appendix B (Table B3).

3.3 Design Assumptions

Several assumptions were made to keep the database focused and manageable, including

- **Order Structure:** Each order belongs to one customer while containing multiple order items.
- **Product Category:** Each product belongs to one category. This simplifies category-based analysis and avoids unnecessary complexity in the product hierarchy.
- **Discount Storage:** The discount is recorded at the order-item level

rather than per unit. This is more suitable for practical promotional cases and avoid rounding issues.

- **Price Snapshot:** unit_price_at_buy and unit_cost_at_sale are stored in the **Order_Items** to record the price and cost at the time of purchase, since these values may change over time.
- **Promotion Validation:** Promotion eligibility and discount calculation occur at the POS or application layer. The system reads the discount type (PERCENT or FIXED), computes the applicable amount, and writes the result as discount_amount_applied into Order_Items, storing only the final outcome, not the logic.
- **Promotion Scope:** The design provides support for both product and category promotions. The product category will not allow order-level promotions, since it would require calculations to be done across different areas of the system.

4. Database Implementation

4.1 Schema Creation & SQLite Data Types

Implemented in SQLite using Data Definition Language (DDL), the schema utilises defensive IF NOT EXISTS clauses to ensure reproducible deployments.

Data type selection aligned with four business requirements: Standardisation, Financial Accuracy, Temporal Integrity, and Operational Constraints. For instance, Financial fields were assigned REAL types for precise ROI calculations, while temporal data utilised ISO 8601 TEXT formatting for chronological auditability. A detailed mapping of attributes to data types and their business rationales is provided in Appendix C (Table C1).

Furthermore, NULL and NOT NULL constraints reflect business reality. Mandatory operational fields (e.g., customer_id) are strictly NOT NULL, whereas fields like promo_id (representing non-promotional transactions) or discount_pct (accommodating mutually exclusive fixed discounts) allow NULL values.

4.2 Database Normalization & Implementation Design Evolution

To transition from the initial conceptual model (Appendix B, Figure B1) to a robust relational database, the schema was strictly normalised to Third Normal Form (3NF) (Connolly and Begg, 2014) to eliminate redundancy and update anomalies. As visualised in the final logical Entity-Relationship (ER) diagram (Appendix B, Figure B2), the design evolved significantly from its conceptual baseline to address three critical analytical and structural challenges:

1. **Resolving Price Ambiguity:**

   The original design utilised a single selling_price field, making it impossible to trace applied discounts. We restructured Order_Items to capture unit_price_at_buy, unit_cost_at_sale, and discount_amount_applied. This preserves immutable historical facts, ensuring precise promotional ROI measurements regardless of future price changes.

   Furthermore, storing discount_amount_applied as an aggregated line-item total prevents systemic rounding errors (e.g., dividing a £100 discount across 3 units), eliminating revenue leakage.

2. **The Profit Calculation Trade-off:**

   In our 3NF design, derived values like final net profit are dynamically calculated rather than stored. In high-volume commercial systems, firms often store calculated profit fields to improve reporting speed. However, the cleaner 3NF structure was deemed more appropriate to prevent update anomalies and demonstrate rigorous relational design principles.

3. **Flexible Promotional Architecture:**

   The M:N relationships between promotions and inventory were resolved via Promo_Product and Promo_Category mapping tables, enabling marketing campaigns to seamlessly target specific items or entire categories.

4.3 Referential Integrity & Business Rule Enforcement

To guarantee functionality, strict constraints were implemented:

1. **Referential Integrity:**

   Primary and Foreign Keys were explicitly defined. Crucially, SQLite's default settings were overridden using PRAGMA foreign_keys = ON to guarantee strict enforcement, preventing orphaned records (e.g., Order_Items must reference a valid order_id).

2. **Financial Protection:**

   CHECK constraints ensure financial metrics (unit_price) cannot be negative. Additionally, CHECK(discount_amount_applied <=

(unit_price_at_buy * quantity)) prevents systemic errors from applying discounts exceeding gross line-item value. UNIQUE constraints on email prevent duplicate registrations.

3. **Cart Aggregation:**

   A UNIQUE(order_id, product_id) constraint in Order_Items enforces cart aggregation, dynamically updating volume rather than creating duplicate lines when products are added multiple times.

4. **System Architecture Trade-offs:**

   Complex cross-table validations like verifying if a loyalty_tier qualifies for a specific promo_id are delegated to the backend Application Logic. SQL TRIGGERS were intentionally avoided to prevent transaction bottlenecks in high-volume environments.

5. Synthetic Data Generation

5.1 Methodology and Volume

To populate the database realistically, a Python script utilising the Faker library was deployed. We adopted a Direct Database Insertion approach, bypassing intermediate CSV files. By inserting directly into SQLite, the engine validated all inputs against CHECK and UNIQUE constraints during creation, ensuring structural compliance. The dataset comfortably exceeds the assignment requirement of 500 records for all main entities (Customer, Product, Orders, and Order_Items). Exact volumetrics and referential integrity test results are detailed in Appendix D (Table D1).

5.2 Data Quality and Realism

The synthetic data was explicitly engineered to be suitable for advanced business analysis. Rather than relying on purely random generation, programmatic business logic was embedded to reflect real retail scenarios. This included implementing dictionary-based shopping cart aggregation to eliminate composite duplicates, skewing promotion redemption rates toward Gold-tier members, and simulating intentional loss-leader clearance sales (Promo P1) that generate realistic negative profit margins for ROI analysis. To ensure analytical validity, several underlying business assumptions, such as baseline pricing margins and chronological ordering, were strictly enforced. A comprehensive mapping of these simulated business rules and assumptions is provided in Appendix D (Table D2).

6. Business Insights and Analysis

The following strategic insights were derived directly from the SQLite database. To ensure transparency and reproducibility, the complete SQL queries encompassing complex aggregations, multi-table joins, and accurate profit calculations are detailed in Appendix E.

6.1 SQL-based Analysis

Promotion analytics include types of promotion, product category, customer segment, and sales channel. SQL queries provided the capabilities for evaluating the effectiveness of promotions.

- **Profit by Promotion:** Profit was calculated as:

  Profit = ((unit_price_at_buy - unit_cost_at_sale) x quantity) - discount_amount_applied.

  This output was grouped by promo_id to compare promotion performance. (Figure E5)

- **Profitability by category:** Order_Items was joined together with Product and Category to evaluate profitability across product categories. (Figure E6).
- **Customer-level analysis:** Promotion effectiveness was analyzed across different customer tiers by linking Order_Items, Orders, Customer, and Membership tables. This allows the comparison of how each loyalty segment responds to promotions (Figure E7).
- **Channel Comparison:** Transactions were grouped by channel to compare the effectiveness of promotions between online and offline sales (Figure E8).

6.2 Visual Evidence and Key Findings

Across the full dataset, non-promoted transactions generated £202,303 in net profit at a consistent ~25% average margin, while all promoted items combined produced a net loss of £14,038. This baseline comparison frames all subsequent findings. Promotions do not necessarily improve overall business performance, even though they can raise the volume of sales. For example, the *"Deep Clean Clearance - 60% Off"* campaign has approximately £22,000 net loss, indicating a large discount could reduce profitability, while moderate promotions perform better. The "Online Flash Sale £10 Off" campaign generated a positive contribution to profit on average margins of 19% (Figure E6). Similarly, the *"Gold Member Exclusive 20%"* campaign was also profitable with a slightly lower impact. This suggests that controlled discounts perform better than excessively high discounts (Ailawadi, Lehmann and Neslin, 2001).

Figure E6 shows that profitability varies by product category. While P3 maintained stable positive margins across most categories, P1 led to a substantial margin loss in Household Cleaning. Customer segmentation further demonstrates that promotional effectiveness differs across tiers. All tiers have negative margins in P1, with Bronze customers showing the largest loss. In contrast, P3 generates positive margins among all, with Bronze customers achieving the highest. Additionally, P2 is only applied to Gold Members with a positive margin, suggesting that targeted promotions for higher-value customers can be more effective. Customized promotions by customer segment are likely to improve profitability.

However, when comparing online and offline channel promotions (Figure E8), there is little difference in profit and margins of trends. Hence, effectiveness depends more on promotion design than the delivery channel.

6.3 Actionable Insights

SmartCart provides actionable insights based on the analyses conducted in this report:

- **Heavy Discounting Risk:** Companies should use heavy discounting carefully, as it may increase sales volume but lead to a significant reduction in profit margins. The first promotion (P1) shows that the use of excessive discounts cannot be sustained as a viable method to generate sales.
- **Moderate Promotion Strategy**: Moderate and targeted promotions should be a priority, as shown in promotions P2 and P3, where sales growth and profitability were achieved through controlled discounting.

- **Category-Based Promotion Strategy**: Profitability varies across categories, discounting should be applied more cautiously in categories where promotions lead to significant margin reductions.
- **Customer segmentation:** Targeting high-value customers, such as Gold members, will make promotions more effective and maximize the return.
- **Channel Impact:** Channel differences are negligible, promotion design drives ROI, not the delivery channel.

6.4 Conclusion

The results show that promotions do not always improve business performance if they are not properly structured and targeted. Organisations should adopt a more balanced promotion strategy rather than relying on heavy discounts. SmartCart should prioritise fixed-amount promotions such as P3, which generate positive returns without structural margin risk, as the preferred mechanism for driving incremental revenue.

References

Ailawadi, K.L., Lehmann, D.R. and Neslin, S.A. (2001). Market Response to a Major Policy Change in the Marketing Mix: Learning from Procter & Gamble's Value Pricing Strategy. Journal of Marketing, [online] 65(1), pp.44–61.

Connolly, T. and Begg, C. (2014) Database Systems: A Practical Approach to Design, Implementation, and Management. 6th edn. Harlow: Pearson.

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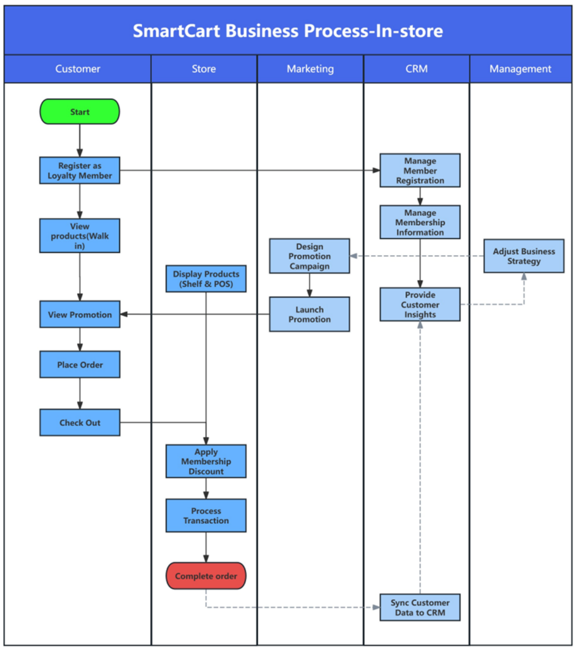
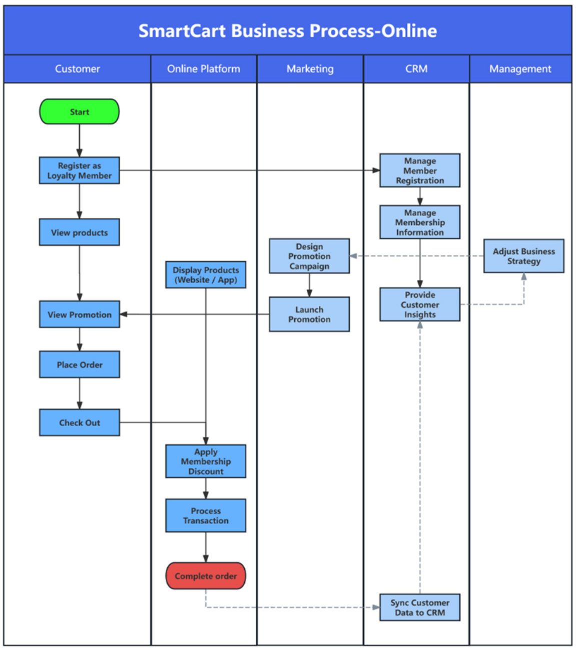
Appendices

Appendix A: Business Process Map





Appendix B:

Figure B1: Conceptual Design before normalization (As-Is)

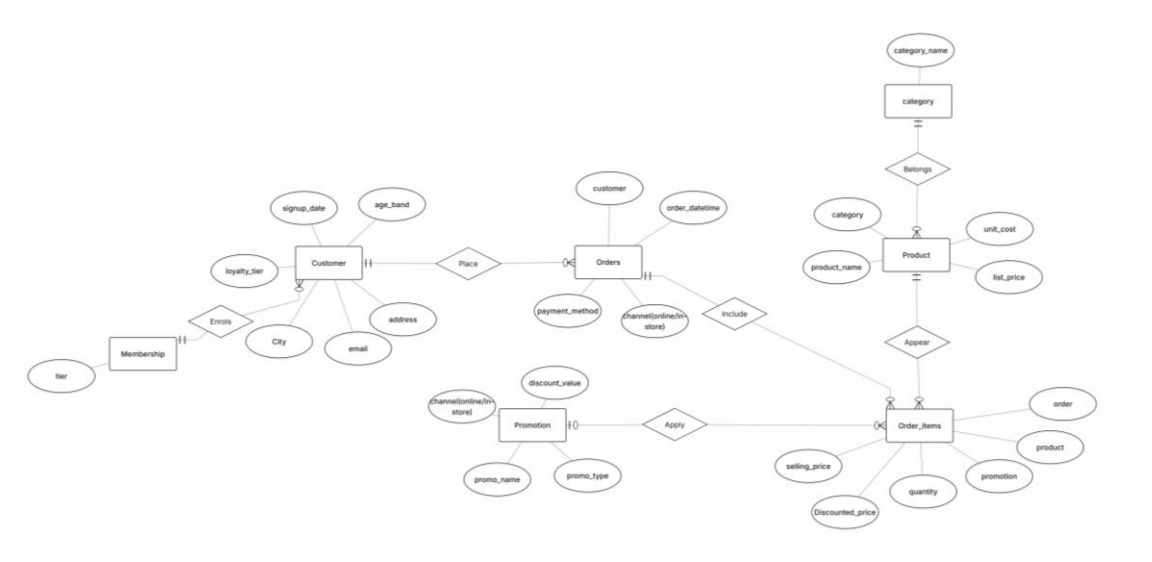


Figure B2: Logical Design after normalization (To-be)

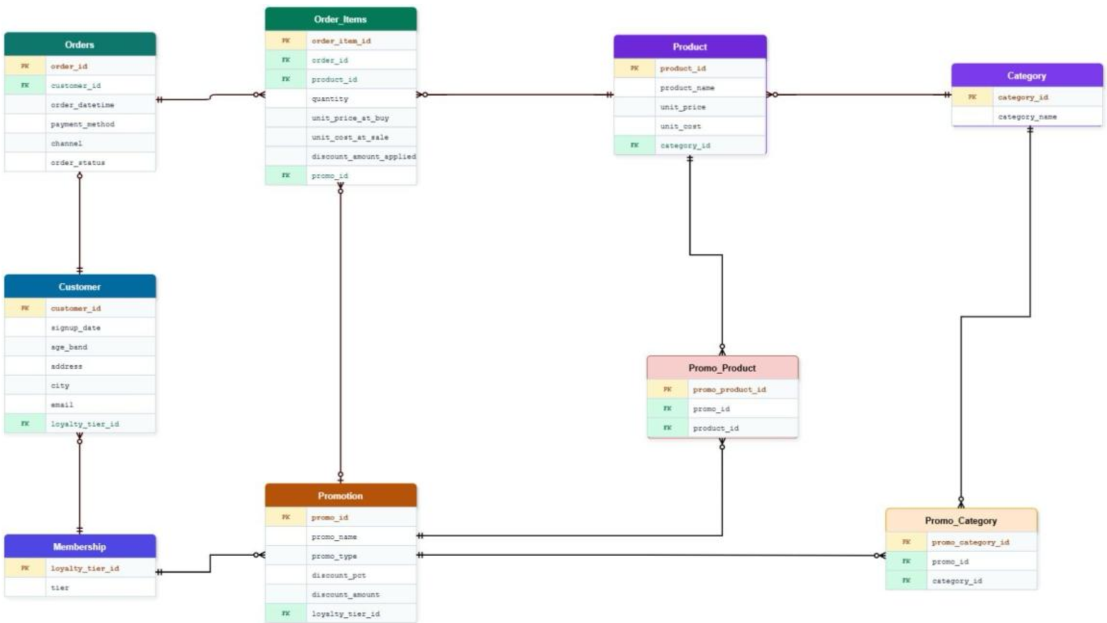


Table B3: Entities summary after normalization (To-be)

Entity Name	Business Purpose	Primary Key (PK)	Foreign Key(s) (FK)
Membership	Defines loyalty tiers (e.g., Bronze, Silver, Gold).	loyalty_tier_id	None
Customer	Stores user profiles and demographic data.	customer_id	loyalty_tier_id
Category	Groups products into broad classifications.	category_id	None
Product	Master inventory tracking and baseline pricing.	product_id	category_id
Orders	Overarching sales transaction headers	order_id	customer_id

	(invoices).		
Order_Items	Granular line-item details within a specific order.	order_item_id	order_id, product_id, promo_id
Promotion	Core marketing campaign rules and discount values.	promo_id	loyalty_tier_id
Promo_Product	Resolves M:N relationship for item-specific promos.	promo_product_id	promo_id, product_id
Promo_Category	Resolves M:N relationship for category-wide promos.	promo_category_id	promo_id, category_id

Appendix C: Database Implementation Details

Table C1: Data types and rationale

Attribute Category	Representative Examples	SQLite Type	Business Rationale
Business Identifiers	customer_id, product_id, order_id	VARCHAR(20)	Standardises key formats across SmartCart's POS and online platforms, ensuring seamless CRM integration.
Financial Fact Data	unit_price_at_buy, discount_amount_applied	REAL	Mandatory for capturing precise floating-point values required to calculate exact net profit and promotional ROI.
Temporal Data	signup_date, order_datetime	TEXT	Tracks customer

			lifecycles and sales trends. ISO 8601 formatting ensures accurate chronological sorting for time-series analysis.
Categorical & Quantitative	quantity, loyalty_tier_id	INTEGER	Enforces whole numbers for discrete data points where fractions are invalid, such as inventory counts and membership ranking.

Appendix D: Synthetic Data Generation Details

Table D1: Main Entity Record Counts and Integrity Validation

Entity Table	Minimum Requirement	Generated Records	Integrity Status
Customer	500	500	0 Nulls in PK, 0 Duplicates
Product	500	500	0 Nulls in PK, Prices > Costs
Orders	500	800	100% attached to valid Customers
Order_Items	500	1,984	Cart aggregated, 0 composite duplicates

Table D2: Data Type Selection Mapping and Rationale

Business Strategy / Scenario	Programmatic Implementation	Analytical Value
"Loss Leader" Clearance	Promo P1 aggressively offers a 60% discount on Household Cleaning items, deliberately exceeding the standard gross product margin.	Generates intentional negative-profit transactions, creating a realistic dataset to evaluate volume-driven ROI.

Customer Churn Analysis	Customer sign-ups were generated with dates preceding the 2025 sales period.	Results in registered but inactive customers, providing a valid cohort for future churn analysis.
Loyalty Program ROI	Gold members were algorithmically assigned an 85% probability of redeeming a promotion, compared to 20% for Bronze.	Emulates the real-world behaviour of engaged customers, validating the effectiveness of the tiered membership strategy.
Chronological Logic	Rule enforced: signup_date strictly precedes order_datetime.	Prevents logical anomalies (e.g., orders placed before account creation), ensuring reliable time-series analysis.
Pricing Strategy (Gross Margin)	Product prices were generated using a strict 20–30% markup multiplier over unit_cost.	Establishes a stable and realistic baseline profitability model, ensuring accurate detection of margin reductions during promotional events.
Channel-Specific Engagement	Online transactions were algorithmically assigned a higher base probability (+15%) of utilizing digital promo codes compared to offline channels.	Reflects the reality of digital marketing behaviors, allowing for cross-channel promotional effectiveness comparisons.

Operational Fulfillment	100% of generated orders possess a 'Completed' status.	Simulates a pre-filtered analytical database strictly focused on revenue-generating transactions, deliberately excluding cancellations.

Appendix E: Data visualization

The following SQL queries were executed to extract the dataset for the business insights and recommendations presented in Section 6:

Query E1: notebook's q1 (Profit by Promotion), matches Figure E5

“SELECT COALESCE(p.promo_name, 'No Promotion') AS promotion, COUNT(*) AS items_sold,

ROUND(SUM((oi.unit_price_at_buy - oi.unit_cost_at_sale) * oi.quantity - oi.discount_amount_applied), 2) AS total_profit

FROM Order_Items oi

LEFT JOIN Promotion p ON oi.promo_id = p.promo_id

GROUP BY promotion

ORDER BY total_profit DESC;”

Query E2: notebook's q2 (Margin by Category × Promo), matches Figure E6

“SELECT c.category_name, COALESCE(p.promo_name, 'No Promo') AS promotion,

ROUND(AVG(((oi.unit_price_at_buy - oi.unit_cost_at_sale) * oi.quantity - oi.discount_amount_applied) / (oi.unit_price_at_buy * oi.quantity) * 100), 2) AS avg_margin_percentage

FROM Order_Items oi

JOIN Product pr ON oi.product_id = pr.product_id

JOIN Category c ON pr.category_id = c.category_id

LEFT JOIN Promotion p ON oi.promo_id = p.promo_id

GROUP BY c.category_name, promotion;”

Query E3: notebook's q3 (Margin by Tier × Promo), matches Figure E7

“SELECT m.tier, COALESCE(p.promo_name, 'No Promo') AS promotion,

ROUND(SUM((oi.unit_price_at_buy - oi.unit_cost_at_sale) * oi.quantity - oi.discount_amount_applied), 2) AS total_profit,

ROUND(AVG(((oi.unit_price_at_buy - oi.unit_cost_at_sale) * oi.quantity - oi.discount_amount_applied) / (oi.unit_price_at_buy * oi.quantity) * 100), 2) AS avg_margin_percentage

FROM Order_Items oi

JOIN Orders o ON oi.order_id = o.order_id

JOIN Customer cu ON o.customer_id = cu.customer_id

JOIN Membership m ON cu.loyalty_tier_id = m.loyalty_tier_id

LEFT JOIN Promotion p ON oi.promo_id = p.promo_id

GROUP BY m.tier, promotion;”

Query E4: notebook's q4 (Margin by Channel × Promo), matches Figure E8

“SELECT o.channel, COALESCE(p.promo_name, 'No Promotion') AS promotion,

ROUND(SUM((oi.unit_price_at_buy - oi.unit_cost_at_sale) * oi.quantity - oi.discount_amount_applied), 2) AS total_profit,

ROUND(AVG(((oi.unit_price_at_buy - oi.unit_cost_at_sale) * oi.quantity - oi.discount_amount_applied) / (oi.unit_price_at_buy * oi.quantity) * 100), 2) AS avg_margin_percent

FROM Order_Items oi

JOIN Orders o ON oi.order_id = o.order_id

LEFT JOIN Promotion p ON oi.promo_id = p.promo_id

GROUP BY o.channel, promotion

ORDER BY o.channel, total_profit DESC;”

Figure E5: Net Profit by Promotion Campaign

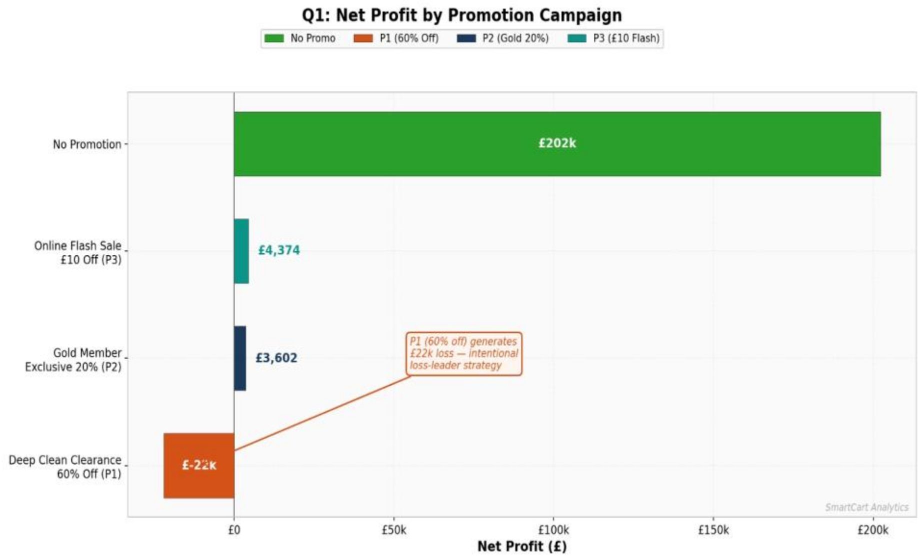


Figure E6: Profit Margin by Category & Promotion Type

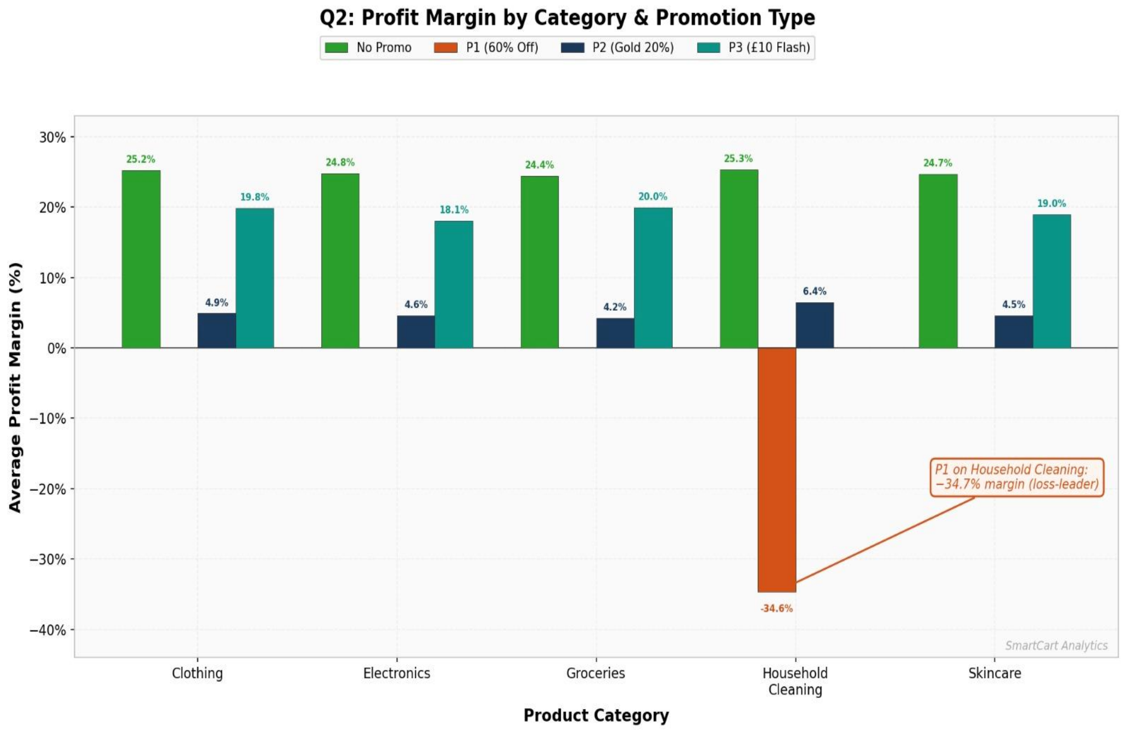
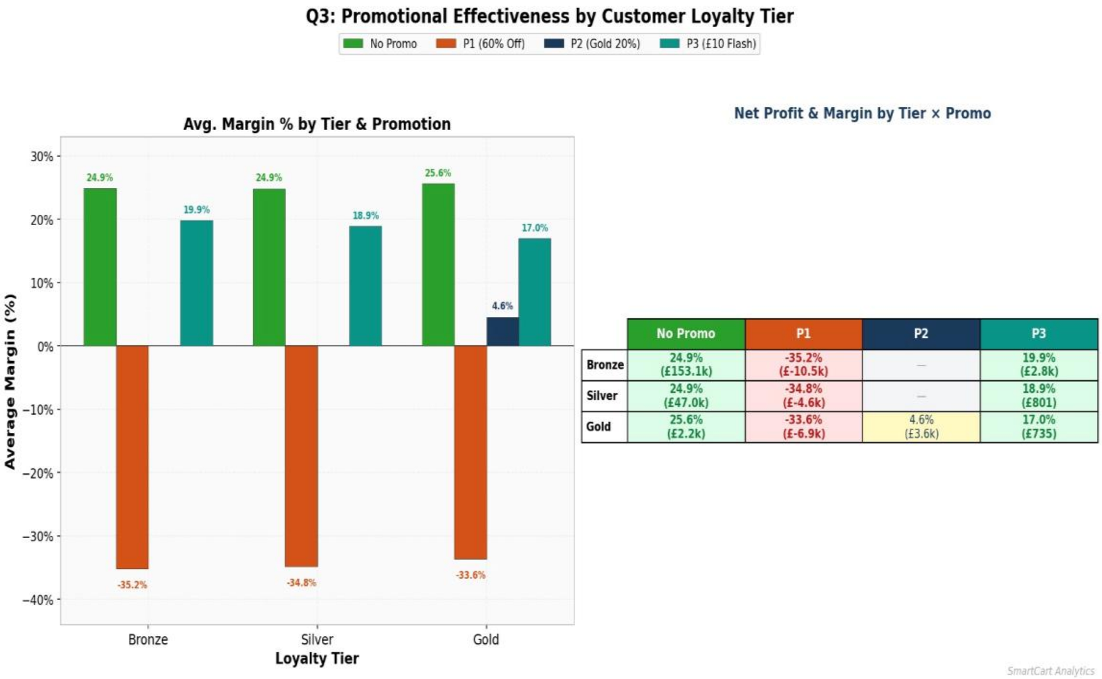


Figure E7: Promotional Effectiveness by Loyalty Tier



	No Promo	P1	P2	P3
Bronze	24.9% (£153.1k)	-35.2% (£-10.5k)	—	19.9% (£2.8k)
Silver	24.9% (£47.0k)	-34.8% (£-4.6k)	—	18.9% (£801)
Gold	25.6% (£2.2k)	-33.6% (£-6.9k)	4.6% (£3.6k)	17.0% (£735)

Figure E8: Promotional Effectiveness by Sales Channel